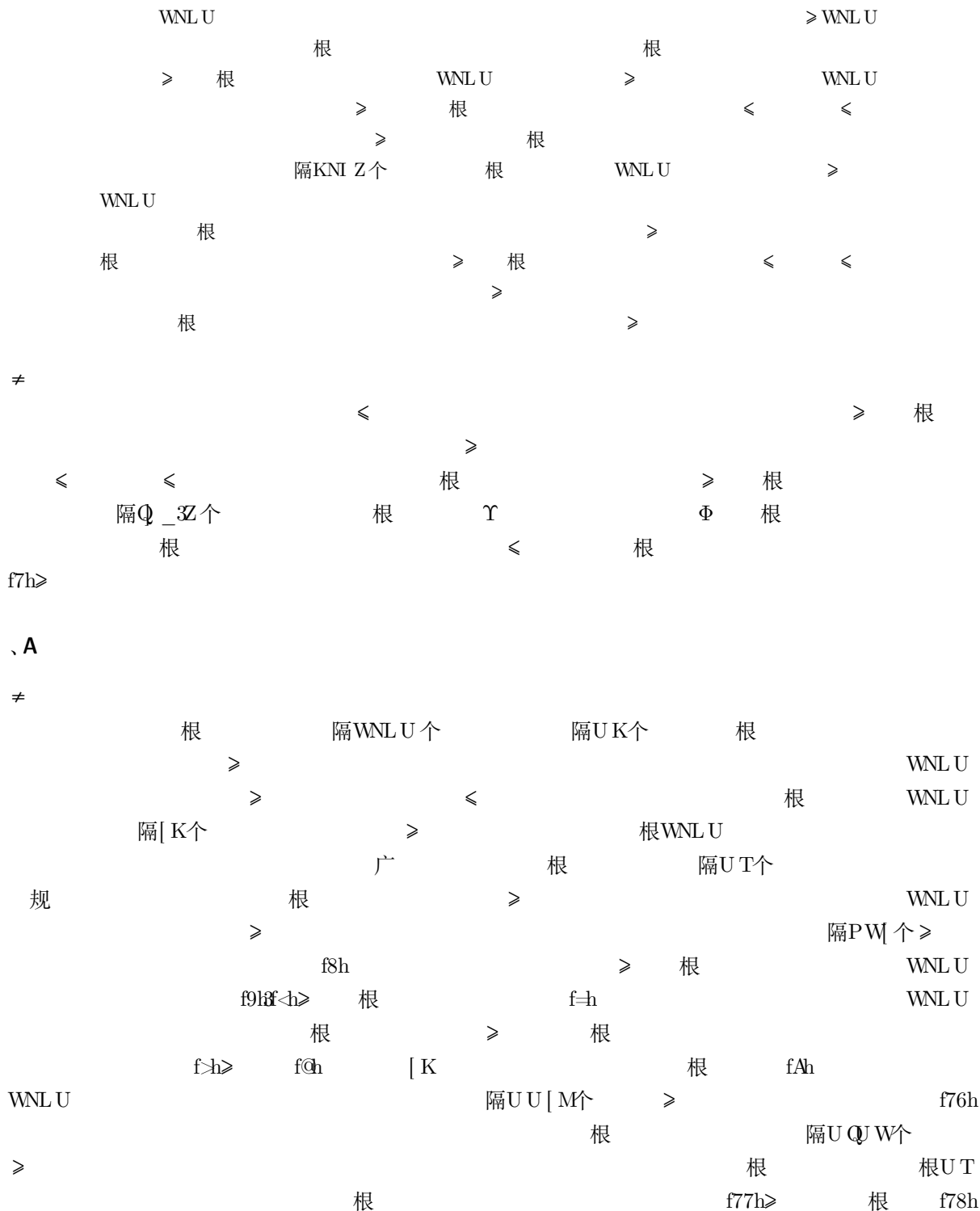


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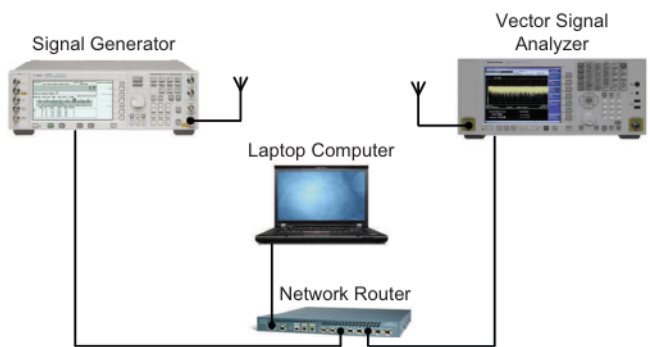
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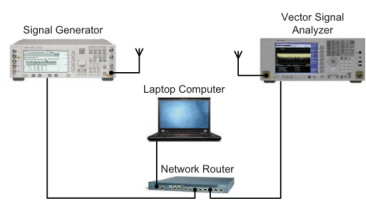
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(a) Measurement setup: transmitter, receiver, laptop computer, router and their connections.

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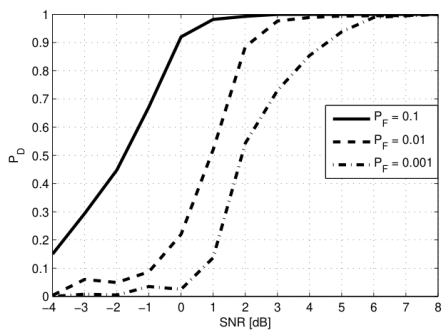


(a) Measurement setup: transmitter, receiver, laptop computer, router and their connections.

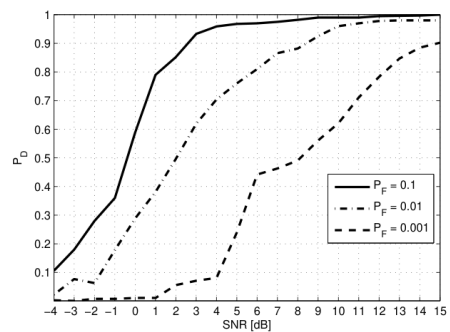


(b) A general overview of the measurement environment.

Fig. 1. The measurement setup and environment.

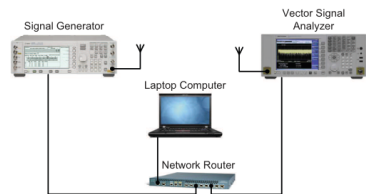


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(a) Measurement setup: transmitter, receiver, laptop computer, router and their connections.



(b) A general overview of the measurement environment.

Fig. 1. The measurement setup and environment.

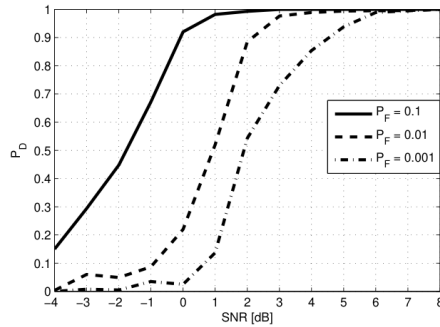


Fig. 3. Probability of detection for SC-FSK vs. OFDM: $N = 1024$.

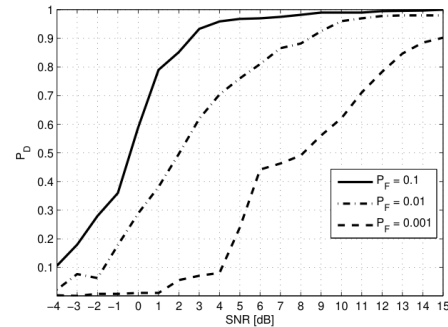


Fig. 4. Probability of detection for SC-QAM vs. OFDM: $N = 10$.

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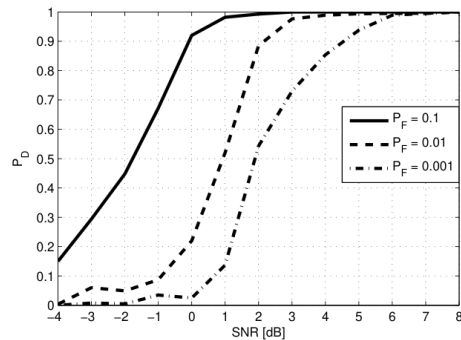


Fig. 2. Probability of detection for SC-FSK vs. OFDM: $N = 1024$, modulation order = 32, $M = 1.5T_S$.

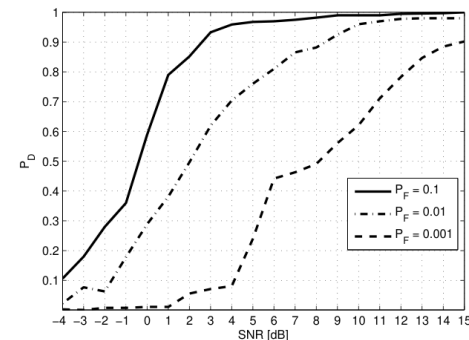
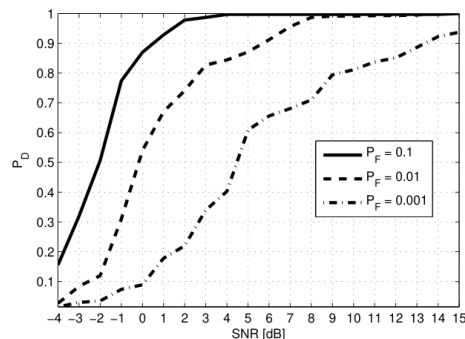


Fig. 4. Probability of detection for SC-QAM vs. OFDM: $N = 10$, modulation order = 32, $M = 1.5T_S$.



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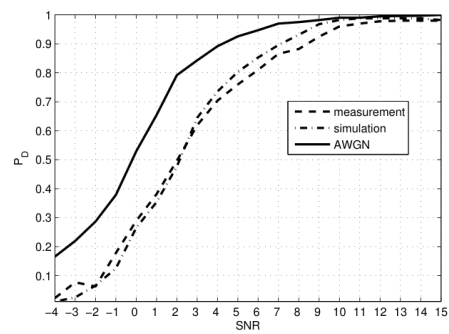
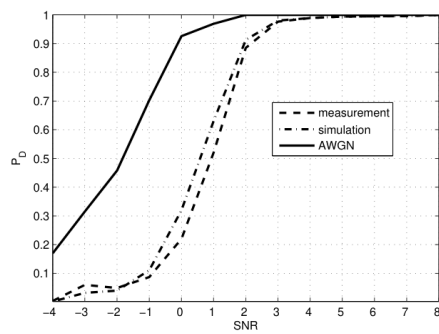
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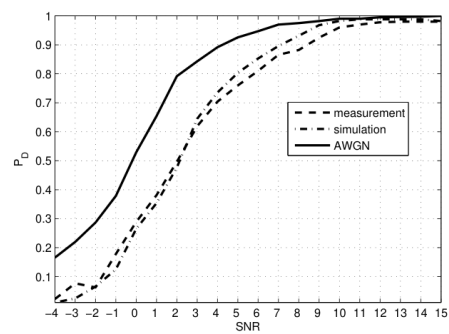
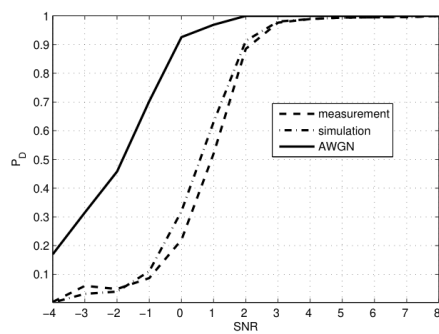
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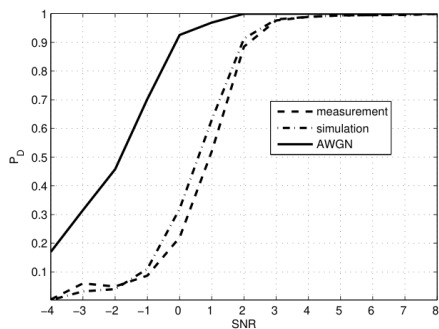


Fig. 5. P_D performance comparison for SC-FSK v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

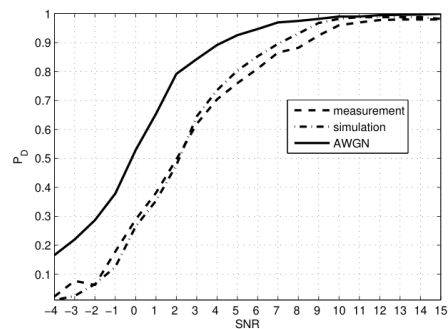


Fig. 7. P_D performance comparison for SC-QAM v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

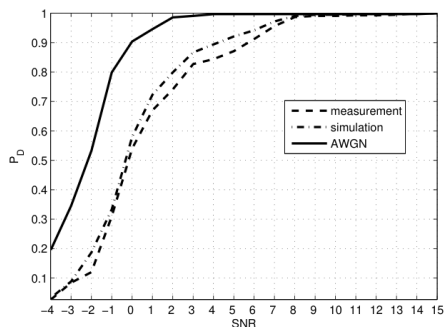


Fig. 6. P_D performance comparison for SC-PSK v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

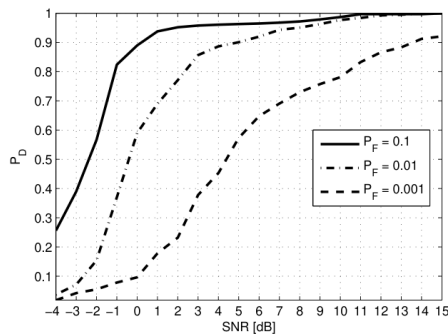


Fig. 7. P_D performance comparison for SC-QAM v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

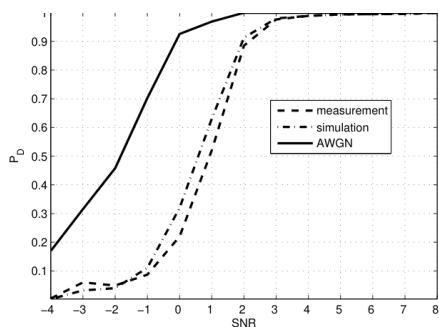


Fig. 5. P_D performance comparison for SC-FSK v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

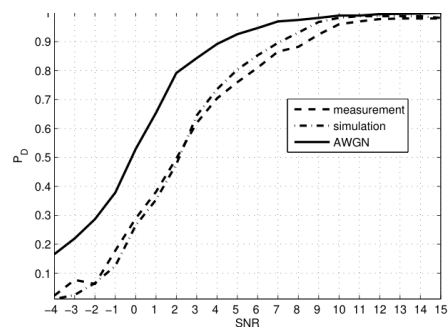


Fig. 7. P_D performance comparison for SC-QAM v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

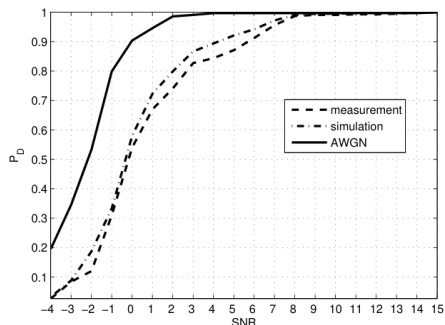


Fig. 6. P_D performance comparison for SC-PSK v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

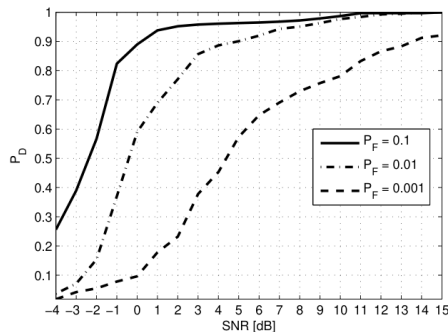


Fig. 7. P_D performance comparison for SC-QAM v.s. OFDM: AWGN channel, simulations, measurements. ($P_F = 0.01$), $N = 1024$, modulation order = 32, $M = 1.5T_S$

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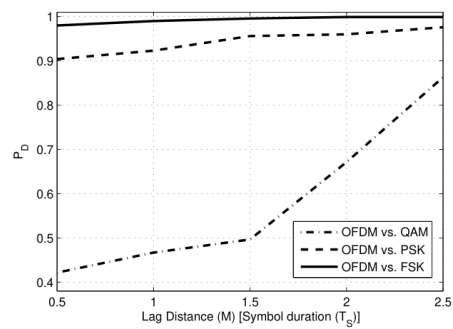
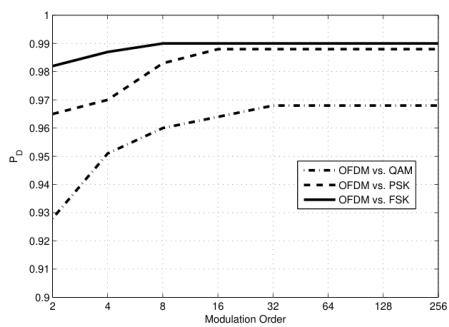
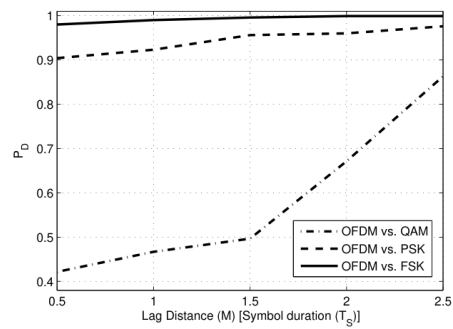
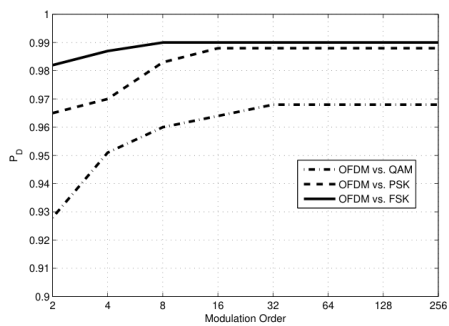


Fig. 11. Effect of lag distance (degrees of freedom) on the identification.

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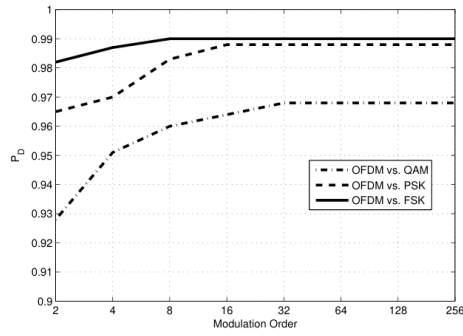


Fig. 9. Effect of modulation order on the identification: ($P_F = 0.1$), SNR = 7 dB, $N = 1024$, $M = 1.5T_S$.

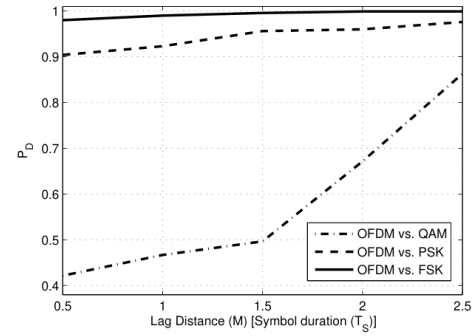
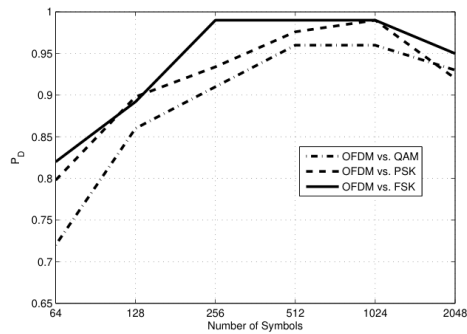


Fig. 11. Effect of lag distance (degree of freedom) on the identification: ($P_F = 0.01$), $N = 1024$, SNR = 7 dB, modulation order = 32.



literature. One of these methods, which is introduced in [15], depends on the fourth and the sixth orders of moments of the OFDM signals and introduces an MMSE algorithm based on these moment functions (MMSE-M). Secondly in [6], a method based on the combination of fourth order moments and the amplitude estimation is introduced (AE-M) and third sums of fourth order cumulants are used for OFDM v.s. S classification in [4] (SUM-C). The coherent method employs a two stage identification procedure to distinguish SLCD, BT-SLCD, and OFDM signals from each other based on the unique second order cyclostationary features of the signals. First the SLCD signals are distinguished from the BT-SLCD/OFDM pair, then these two signals are identified in the second stage [16] (BT-SLCD), however the first stage of the signal classification algorithm is investigated for the

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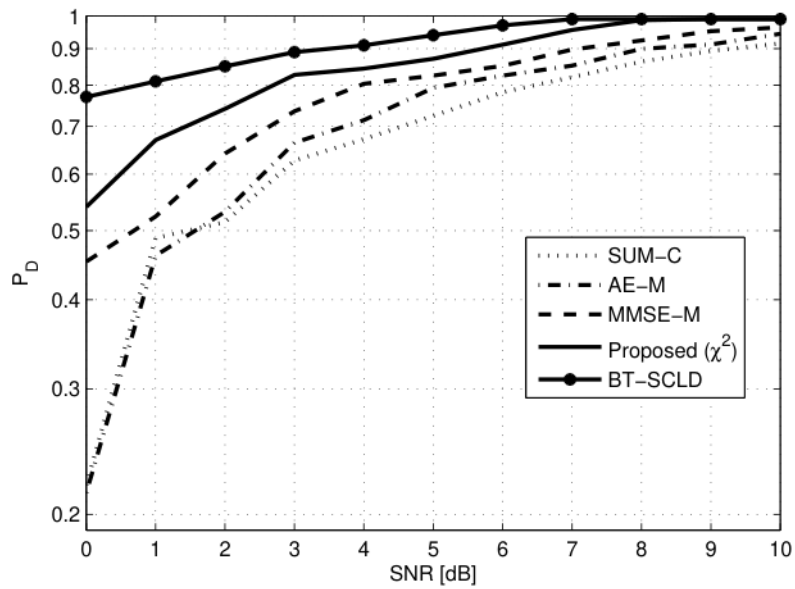
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f7h Q _ Zlotnzz x .ytnl,tzy | pn,z}2 | pnsytnl wQpy,tqnl,tzy zq Ltrt,lw | tryl w2 Q _Z [U 47=66372 | p{ ,px np} 86784

f8h I 4 | -1x t2 l yo J 4 U 4 | l owp}2 Ptp}l }nstn wLtrt,lw Uzo.w,tzy Kw tqnl,tzy _ tyr K.x .wy, 2 Q QMM | }ly ln,tzy zy Kzx x .ytnl,tzy 2 zw: @yz492{{4: 7=3 8A2U1} 86664

f9h L4 O}tx l wt2 | 4 Z1{.lyz2 l yo T4 Lp at,z 2 I y I .,zx l,tn Ltrt,lw Uzo.w,tzy Kw tqp} q} Upl .}px py, zy | pwnzx x .ytnl,tzy Vp,-z}v '2 Q QMM | }ly ln,tzy zy Q ,}x py,l,tzy l yo Upl .}px py,2 zw <=2 yz4 <2 {{4 7>737>862 866>4

f: h U 4 { st2 I 4 Tl .o}p}2 d 4 J1 }3Vp 2 l yo b 4 | .2 Nz .}s W}op} K.x .wy, ty Lt ,tyr .t sty r | tyrp Kl }tp} qjzx WNL U | tryl w2 Q X}zn4 QMM U tw,l }" Kzx x .ytnl,tzy Kzyq}pyup .U QKWU 866@2 | l y Ltr z2 Kl 2 _4 4 47=37A Vz -4866@2 {{4 73=

f-h b 4 Tt.2 R4 b l yr2 | 4 Tt2 J wyo Lp,pn,tzy l yo M ,tx l,tzy zq WNL U | tryl w ty Kzryt,t p Zlotz Kzy,p " 2 Q X}zn4 8yo Q ,p}yl ,tzyl wKzyq}pyup zy | tryl wX}znp tyr | " ,px .QK[X] 8676/2 L l w l y2 Kstyl 2 <3> R.w 86762 {{49: >30<74

f-h S 4 _w pn2 Zpnzryt,tzy zq WNL U Uzo.w,tzy Up,szo2 Q R. .}yl wzq Zlotzpyrtypp}tyr2 zw 7>2 yz4 72 {{4 <63<2 I { }tw 866@4

f-h P4 b l yr2 J 4 Tt2 l yo d 4 b l yr2 Uzo.w,tzy Qpy,tqnl,tzy q} WNL U ty U .wt{l ,s Kt}n.x ,l ynp J1 po zy | .{{z} ,a pn,z } U1nstyp2 Q X}zn4 77,s QMM | tryl {z}p Q ,p}yl ,tzyl wKzyq}pyup zy Kzx x .ytnl,tzy | " ,px .QK[866@2 O .l yr sz .2 Kstyl 27A37 Vz -4866@{{479: A379<9

f@h b pt L1t2 d 4 b l yr2 R4 b l yr' Rty, Xz-p} M ,tx l,tzy l yo Uzo.w,tzy Kw tqnl,tzy _ tyr | pnzyo l yo Ptrsp} | ,l,t ,tn '2 Q X}zn4 QMM b t}pp Kzx x .ytnl,tzy l yo Vp,-z}vtyr Kzyq}pyup .b KVK8668/2 zw 72 W}wyo z2 Nw}tol2 _4 4 47>387 U1} 86682{{47<37<@

fAh J ty b l yr2 Ttyozyr Op2 J wyo Qpy,tqnl,tzy zq WNL U | tryl wty Z1" wptrs Kslyypw2 Q X}zn4 QMM <s Q ,p}yl ,tzyl w Kzyq}pyup zy Qq{x l,tzy2 Kzx x .ytnl,tzy l yo | tryl wX}znp tyr2 J1 yr vzv2 | sl twyo2=3A Lpn4866<2 {{4A-63A<

f76h U 4 U .s w l . 2 U pyr .n Wyp}2 W4 Lzn}p2 l yo N4 Ryo}l w I w-nzx {w t," x zo.w,tzy mw tqnl,tzy lwz}t,sx q} U Q W " ,px 2 Q QMM Kzx x .ytnl,tzy Tp.,p} 2 zw 7>2 yz4762{{47@@737@@ 286794

f77h R4 Tptyzy py2 l yo U 4 R.y.,t2 Uzo.w,tzy Kw tqnl,tzy ty I ol { ,t p WNL U | " ,px 2 Q X}zn4 QMM <As a ptn.w } | pnsyzwr" Kzyq}pyup .a | K 866: 3 { }tyr/2 zw 2 U twy2 Q l w 27>37A U1" 866: 2 {{47<< 37<<@

f78h N4 b l yr2 l yo J 4 Tt2 I Vp- Up,szo q} Uzo.w,tzy Kw tqnl,tzy J1 po zy Jzz, ,jl { | pnsyt} .p2 Q X}zn4 QMM Q ,p}yl ,tzyl w| "x {z t.x zy Kzx { .,p} | ntpynp l yo Kzx { .,l ,tzyl w| pnsyzwr" .QK[K] -6@/2 zw 82 | sl yrs l t2 Kstyl 2 86388 Lpn4866@{{47737:

f79h R4 eslyr2 l yo J 4 Tt2 I Vp-Uzo.w,tzy Qpy,tqnl,tzy [nsp x p q} WNL U ty U .wt{l ,s Z1" wptrs Nlotyr Kslyypw2 Q X}zn4 QMM Q ,p}yl ,tzyl w| "x {z t.x zy Kzx { .,p} | ntpynp l yo Kzx { .,l ,tzyl w| pnsyzwr" .QK[K] -6@/2 zw 72 | sl yrs l t2 Kstyl 286388 Lpn4866@{{4>A03A=

f7: h S 4V4 P1 | 2 I 4 Uly z .}2 | 4 Vz }osz w 2 Kw tqnl,tzy zq Ltrt,lw Uzo.w ,po | tryl w J1 po zy | tx p N}p| .pyn" Zp{p py,l,tzy2 Q X}zn4 8yo Q ,p}yl ,tzyl wKzyq}pyup zy | tryl wX}znp tyr | " ,px .QK[X] 8676/2 L l w l y2 Kstyl 2 <3> R.w 86762 {{473<

f7-ch I 4 X.ynsts p-1 2 W4 4 Lzn}p2 | 4 Z1 d y2 l yo Z4 Qvz w K"mw ,l ,tzyl }t," 3nl po I wz}t,sx q} J wyo Zpnzryt,tzy zq WNL U l yo | tyrp Kl }tp} Ttypl | Ltrt,lw Uzo.w,tzy 2 Q X}zn4 QMM 7@s Q ,p}yl ,tzyl w| "x {z t.x zy Xp} zyl w Qozz} l yo U zntwp Zlotz Kzx x .ytnl,tzy .XQZK 866>2 I ,spy 2 O}ppnp293> | p{ ,4866>2 {{473<

f7-h Yt".y eslyr2 Wh,l d I 4 Lzn}p2 d l st I 4 Mopx p}ol s2 | 4 Z1 d y2 l yo Z4 Qvz w | pnzyo3W}op} K"mw ,l ,tzyl }t," zq J | 3 KTL | tryl wB | spz}p,tnl wLp pw{x py, l yo I {{vtnl,tzy ,z | tryl wKw tqnl,tzy l yo J wyo Xl }l x p,p} M ,tx l,tzy2 Q QMM | }ly ln,tzy zy b t}pp Kzx x .ytnl,tzy 2 zw 782 yz4: 2 {{47<6737<77286794

f7>h I w I w P1 nl syl2 Wh,l d I 4 Lzn}p2 Z1 x l ns l yo }ly a pyvl ,p l y2 l yo Ltx t,}tp K4 Xz{p n.2 | pnzyo3}op} n"mw ,l ,tzyl }t," zq x zntwp b tU I c l yo Tl M WNL U tryl w l yo l {{vtnl,tzy ,z {pn.}x l-1}pyp ty nzryt,t p }lotz " ,px 2 Q QMM R. .}yl wzq | pwpn,p o | z{tn ty | tryl wX}znp tyr2 zw =2 yz472 {{48=3 8286784

f7@h I 4 J z . xpr x2 X4 Ktmw ,2 X4 R wzy2 Vp-I wz}t,sx q} J wyo Zpnzryt,tzy zq WNL U J1 po | " ,px '2 Q Mwp tp} | tryl w X}znp tyr R. .}yl w zw A62 yz492 {{4A663A79286764

f7Ah I w l }nt P1 x 2 U zntpy | szl tn2 | l wps I 4 I wsnptw2 l yo I w .}vt Nl slo2 I y z p} tp-zq ql .,p 3nl po x p,szo q} otrt,lw x zo.w,tzy mw tqnl,tzy2 Q X}zn4 QMM 7 , Q ,p}yl ,tzyl wKzyq}pyup zy Kzx x .ytnl,tzy 2 | tryl wX}znp tyr2 l yo ,spt} I {{vtnl,tzy .QK[XI -79/2 | sl }d s2 _I M27837: Npn486792 {{473-4

f86h P4 Tt2 d 4 J1 }3Vp 2 I 4 I not2 W4 4 | zx pvs2 l yo b 4 | .2 WNL U Uzo.w,tzy Kw tqnl,tzy l yo Xl }l x p,p} M ;}l n,tzy2 Q X}zn4 QMM 7 ,